INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Business Continuity and Disaster Recovery Assignment

GRC205 Incident Management and Business Continuity Week 21

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Task 1 Business Impact Analysis (BIA)

- Build a BIA covering F1 to F5. For each function, record the owner role, users affected, critical resources, dependencies, impact over time, minimum acceptable service, and proposed recovery priority. Include people, premises, suppliers, and information needs.

ANSWER

Five Function BIA

Table 1. Five-function Business Impact Analysis (F1 to F5)

Function	Owner role and users affected	Critical resources and dependencies	Impact over time (no recovery)	Minimum acceptable service and controls	Priority
F1 Teaching and assessment	Head of Academic Services; Examinations Officer. 1,800 learners; 120 candidates sitting the 11:00 examination; teaching staff.	LMS01, DB01, AUTH01, recovery network; M365 for email and Teams; invigilators; 25 alternate workstations from 10:00.	0 to 3 h: teaching disrupted, candidates anxious. 3 h (11:00): examination fails and NGN 120,000 rescheduling cost is triggered. 8 h+: lost teaching day, integrity and welfare complaints.	Examination delivered at 11:00 through an approved alternative (Task 4) with invigilation, attendance register and secure script handling. Teaching continues through M365 Teams.	1
F2 Fees and finance	Finance Manager.	FIN01, DB01, AUTH01; external payment provider	Receipts and wallets cannot be updated from 08:00. Fee	Payments continue through the provider. Provisional receipt	2

	Finance officers; learners paying fees; sponsors.	portal and reports; M365 for temporary register.	confirmation due 16:00 (8 h). After 16:00 learners may be wrongly shown as unpaid.	register in a restricted M365 workbook; preparer and approver separated; daily provider report retained for reconciliation.	
F3 Admissions and registration	Admissions Manager. Applicants; registration staff; existing learners changing courses.	REG01, DB01, AUTH01; M365 Forms and shared mailbox.	New applications and changes cannot be entered. Admissions close 18:00 (10 h); late applicants may be lost.	Applications captured on a controlled M365 form with time stamp and reference number; no offers issued until REG01 entry is checked.	3
F4 Staff payroll	HR Manager with Finance Manager. 90 staff.	HR01, DB01, AUTH01; bank submission channel; last approved payroll file.	No immediate impact on Day 0. Bank deadline 12:00 Day 1 (28 h). Wrong or duplicate instructions affect staff and need manual correction.	One approved payroll batch only, with unique batch reference, dual authorisation and bank confirmation check. No manual changes without HR and Finance approval.	4
F5 Records and certificates	Registry and Records Manager. Graduates, employers, awarding partners.	FS01, AUTH01, recovery network; access-controlled storage.	Requests queue from 08:00. 48-hour service commitment ends 08:00 Day 2. Integrity and confidentiality failures would damage trust.	Requests logged with reference and due time. Certificates issued only from restored and verified FS01 files with existing access restrictions.	5

Table 2. People, premises, supplier and information needs

Need	Requirement during the 72-hour outage	Functions
People	Incident Commander and deputy; BC/DR Coordinator; two technicians (T1, T2) for all recovery actions (A16); at least six invigilators for two rooms; finance preparer and approver; admissions officer; HR officer; registry officer; communications lead; learner support staff.	All
Premises	Main site unavailable for 72 hours. Alternate facility with 25 workstations from 10:00 and assumed seating for 40 candidates (A12). Remote working for other staff through M365.	F1 (critical), all
Suppliers	Alternate facility provider; offsite backup provider; warm recovery hosting; payment provider; bank; mobile internet carriers; Microsoft 365.	All, especially F1, F2, F4
Information	Offsite backup catalogue (B01 to B04 and the disconnected full); examination papers and candidate list; payment provider reports; last approved payroll file; contact lists held off the main site; this plan in printed and M365 copies.	All

- Calculate the total estimated disruption cost for 8 hours and 24 hours using the supplied rates and the one-off examination trigger. Show the calculation for each function and the combined total. Explain two limitations of these estimates and distinguish financial impact from learner welfare, reputation, and data integrity.

ANSWER

The brief instructs that hourly amounts are separate, non-overlapping incremental costs, applied for the first 24 hours, with the NGN 120,000 rescheduling cost added once when its trigger is met. With no recovery or continuity measure, no acceptable assessment service exists at 11:00, three hours into the outage, so the trigger is met within both periods. The combined hourly rate is NGN 135,000 (60,000 + 40,000 + 20,000 + 10,000 + 5,000).

Table 3. Estimated disruption cost with no recovery or continuity measures (exercise estimates, not observed losses)

Function	Rate (NGN/h)	8-hour working	8-hour impact (NGN)	24-hour working	24-hour impact (NGN)
F1	60,000	60,000 × 8 = 480,000; + 120,000 rescheduling (trigger at 11:00)	600,000	60,000 × 24 = 1,440,000; + 120,000	1,560,000
F2	40,000	40,000 × 8	320,000	40,000 × 24	960,000
F3	20,000	20,000 × 8	160,000	20,000 × 24	480,000
F4	10,000	10,000 × 8	80,000	10,000 × 24	240,000
F5	5,000	5,000 × 8	40,000	5,000 × 24	120,000
Total	135,000	135,000 × 8 = 1,080,000; + 120,000	1,200,000	135,000 × 24 = 3,240,000; + 120,000	3,360,000

The total estimated disruption cost is therefore NGN 1,200,000 at 8 hours and NGN 3,360,000 at 24 hours. These are exercise estimates and not observed losses. F1 accounts for half of the 8-hour total because of its high rate and the one-off examination cost.

Two limitations matter. First, a flat hourly rate ignores timing: an hour lost at 10:30 before the examination is worth far more than an hour lost at 02:00, and the brief stops costing at 24 hours although the site is closed for 72, so the true cost of the full event is understated (A2). Second, the rates are averages with no evidence base or range. They exclude recovery costs such as overtime, facility hire and reconciliation effort, and they assume impacts do not interact, although a failed fee confirmation could also increase admissions complaints.

Financial impact must be kept separate from three non-financial impacts. Learner welfare covers candidates travelling through flooded areas, stress from an uncertain examination and fair treatment of

learners with access needs. Reputation covers trust from learners, sponsors and awarding partners, which may fall faster than any hourly rate suggests. Data integrity covers wrong fee balances, duplicate payroll and altered certificates, which can cause harm long after the site reopens. A low-cost function such as F5 can therefore still carry high integrity risk.

- Explain how the 11:00 examination and later deadlines affect priorities. Separate business priority from the technical restoration order required by shared infrastructure. Identify at least three important dependency risks.

ANSWER

Priorities and dependency risks

Business priority follows deadline and impact. The 11:00 examination is the earliest and costliest deadline, so F1 is first. F2 follows because fee confirmation is due at 16:00 and errors affect learners directly. F3 is third with an 18:00 close. F4 has until 12:00 on Day 1, and F5 has a 48-hour commitment and the lowest rate (ICDFA, n.d., slide 5).

Technical restoration order is different, because shared infrastructure must come first whatever the business priority. No function can be restored before the recovery network, AUTH01 and the security and backup checks, and F1 to F4 all wait for DB01. F5 is the lowest business priority but can be recovered first, because FS01 needs only AUTH01 and the checks and can run on the second technician while DB01 restores. Table 4 shows the resulting sequence.

Table 4. Justified technical restoration sequence

Order	Technical step	Why it comes here	Business functions released
1	Recovery network	No prerequisite; every action needs it.	None directly
2	AUTH01 restoration	Local applications authenticate against AUTH01.	None directly
3	Security and backup checks	Confirms backups are intact before data is trusted.	None directly
4a	DB01 restoration (T1)	Single shared database for F1 to F4.	Enables F1 to F4
4b	FS01 restoration (T2, parallel)	Independent of DB01; uses the idle technician.	F5
5	LMS01, then FIN01, REG01, HR01	Applications in business-priority order.	F1, F2, F3, F4

Four dependency risks are most important:

- **DB01 single point of failure.** One database supports four functions. If the B02 incremental chain fails, F1 to F4 all lose data or time together.

- **Shared administrator identity.** Production and backup administration use one identity, so one compromised or misused credential could alter both live systems and their backups.
- **Alternate facility capacity.** Only 25 workstations exist for 40 candidates who need one, and they are not available until 10:00, one hour before the examination.
- **Offsite provider access independence.** The provider holds the encrypted copies, but its access-independence controls are undocumented, so it cannot yet be shown that the copies are beyond reach of the same compromise.

Recovery objectives and strategy selection

Task 2 Recovery objectives and backup design

- Define Recovery Time Objective (RTO), Recovery Point Objective (RPO), and Maximum Tolerable Downtime (MTD). Propose and justify numerical targets for each function. State the service level at which you will declare recovery and leave a justified buffer between technical recovery and the tolerable business disruption limit.

ANSWER

Definitions and objective matrix

- **The Recovery Time Objective (RTO)** is the target time from disruption to restoring a function at its declared service level.
- **The Recovery Point Objective (RPO)** is the maximum acceptable data loss, measured as time back from the disruption to the last recoverable data point.
- **The Maximum Tolerable Downtime (MTD)** is the longest outage the institute can accept before harm becomes unacceptable.

MTD as encompasses the RTO plus additional recovery and validation time. Every RTO below is set strictly less than its MTD, and the gap is a deliberate buffer for validation, reconciliation and the chance that a step fails.

Recovery is declared at business acceptance, not when a server starts. The declaration level for each function in Table 5 is the test that its owner signs. F1 is split because the examination sitting has a much shorter tolerance than routine teaching. All targets are my proposals pending sign-off by function owners (A4).

Table 5. Recovery objective matrix (RTO, RPO and MTD) with declaration levels

Function	Recovery declaration level (owner signs)	RTO	MTD	Buffer	RPO	Justification
F1a Examination sitting	Approved alternative delivery running for all 120 candidates with invigilation.	Continuity by 10:45 (2 h 45)	3 h (11:00)	15 min	n/a (new data)	Examination fixed at 11:00; rescheduling cost and welfare harm follow.

F1b LMS teaching	Learners and staff log in, open courses and save a test submission, verified by Examinations Officer.	4 h (12:00)	8 h (16:00)	4 h	1 h	Loss of a whole teaching day is not tolerable; submissions change hourly.
F2 Fees and finance	FIN01 posts receipts and 10 sampled provider payments reconcile.	5 h (13:00)	8 h (16:00)	3 h	1 h	Fee confirmation due 16:00; reconciliation needs at least 2 to 3 h.
F3 Admissions	New application entered and retrieved; form backlog loading started.	6 h (14:00)	10 h (18:00)	4 h	2 h	Admissions close 18:00; applications also captured on forms.
F4 Payroll	HR01 produces payroll file that matches last approved totals and is approved by HR and Finance.	24 h (08:00 Day 1)	28 h (12:00 Day 1)	4 h	4 h	Bank deadline; payroll data changes slowly.
F5 Records	Sample certificates open, checksums match, access restrictions confirmed.	24 h (08:00 Day 1)	48 h	24 h	4 h	48-hour commitment; files created during the day must not be lost.

The tightest buffers are F1a and F2. They are acceptable only because both rely on continuity workarounds that do not depend on technical restoration.

- Calculate the age of the latest recoverable DB01, AUTH01, and FS01 checkpoints at 08:00. Map the resulting potential data-loss windows to the affected functions and compare them with your proposed RPOs. Use the disruption time as the reference point. Explain what can be reconciled from temporary records and what cannot be assumed recoverable.

ANSWER

Checkpoints and data loss windows

Checkpoint age is measured from the 08:00 disruption back to the latest recoverable point, because that is the data at risk, as Table 6 shows.

Table 6. Checkpoint age at 08:00 and comparison with proposed RPOs

System	Latest recoverable checkpoint	Working (to 08:00 Day 0)	Age (data-loss window)	Functions affected	Proposed RPO	Gap
DB01	B01 full 23:00 Day −1 plus every B02 incremental to 07:00 Day 0	08:00 − 07:00	1 h	F1b, F2, F3, F4	1 h (F1b, F2); 2 h (F3); 4 h (F4)	Met, with no margin for F1b and F2
AUTH01	B03 snapshot 00:00 Day 0	08:00 − 00:00	8 h	All (identity and configuration)	4 h	Exceeds by 4 h

FS01	B04 full 23:00 Day −1	1 h (Day −1) + 8 h (Day 0)	9 h	F5	4 h	Exceeds by 5 h
Fallback	Disconnected full 17:00 Day −3	7 h (Day −3) + 24 h + 24 h + 8 h	63 h	All	All	Exceeds every RPO

The DB01 chain must be complete. B01 alone gives a 9-hour window, and any missing or corrupt incremental breaks every later link. The disconnected full is the only copy proven to be offline, so it is the fallback if the online chain cannot be trusted. Day 0 is a Monday, so the 63-hour window spans a weekend. Activity may have been low (A8), but that is unverified.

Some lost data can be rebuilt from temporary or external records: payments from the provider's reports, applications received by email or form, payroll from the last approved file and bank records, and certificate requests from M365 mail. Other data cannot be assumed recoverable: LMS submissions and marks entered between 07:00 and 08:00, AUTH01 password resets and account changes after 00:00, FS01 files saved after 23:00, and any transaction inside a failed incremental (A6). These must be identified with users and re-entered, not presumed present.

- Design a backup schedule that addresses the RPO gaps. Specify full and incremental or differential backups, retention, offsite placement, and the exact chain needed to restore DB01. Apply the 3-2-1 principle and include an independent offline or immutable copy, restricted access, encryption, and restoration tests.

ANSWER

Backup and restoration design

Table 7. Proposed backup schedule and retention

System	Full	Incremental or differential	Achieved RPO	Retention
DB01	Weekly full, Sunday 23:00	Daily differential 23:00; incremental every 30 minutes, 06:00 to 22:00	30 min (working day)	Incrementals 14 days; differentials 35 days; weekly fulls 13 weeks; monthly offline 12 months
AUTH01	Daily snapshot 00:00	Additional snapshots 08:00, 12:00, 16:00, 20:00	4 h	35 days; monthly offline 12 months
FS01	Weekly full, Sunday 23:00	Incremental every 2 hours, 06:00 to 22:00, and 23:00 daily	2 h	Incrementals 35 days; fulls 13 weeks; monthly offline 7 years for certificates (A9)
LMS01, FIN01,	Weekly image	After every approved change	Configuration only	13 weeks

REG01, HR01				

Table 8. 3-2-1 placement with an independent immutable and offline copy

Copy	Location and media	Protection	Access
1 Production	Primary or warm recovery systems	Normal controls	System administrators
2 Local backup	Backup appliance at the warm recovery site (disk), away from the flood-affected site	Encrypted; integrity hashes	Separate backup administrator identity with MFA
3 Offsite immutable	Offsite provider object storage with object lock for 35 days (cloud)	Encrypted with Northbridge-held keys (A7); deletion blocked during lock	Provider write-only role; restore needs two named Northbridge approvers
4 Offline	Monthly full of DB01, AUTH01, FS01 to encrypted removable media, stored in a fire and flood-rated safe off the main site	Disconnected after writing; checked quarterly	Two-person retrieval log

- Explain why offsite storage or a live replica alone does not establish recoverability. Define how you will check backup integrity, access independence, and the usability of restored data. Label any unverified assumptions.

ANSWER

Exact DB01 restore chain. For this event: B01 full (23:00 Day −1), then every B02 incremental in catalogue sequence up to and including the 07:00 file. I assume these are hourly files from 00:00 to 07:00 (A5). Under the new design the chain is the latest weekly full, then the latest daily differential, then each 30-minute incremental since that differential, in order, with the hash of each link verified before applying the next.

Access and encryption. Production and backup administration will use separate identities with phishing-resistant MFA, no shared accounts and a sealed break-glass account. The provider must document in contract how its staff and Northbridge's production credentials are prevented from deleting or changing protected copies. Until then, the independence of copy 3 is an unverified assumption.

Offsite storage proves only that a copy exists elsewhere. It does not prove that the copy is complete, uncorrupted, decryptable or restorable within the RTO. A live replica copies changes almost instantly, so corruption, accidental deletion or malicious changes reach it too. Recoverability is shown only by restoring.

Table 9. Backup verification controls

Check	Method	Frequency	Evidence

Backup integrity	Automated hash and catalogue-chain check for every job; alert on any missing or failed link	Every job	Daily integrity report
Access independence	Attempt deletion of a locked test object using production credentials (should fail); review provider access logs	Quarterly	Signed test record
Restore and usability	Restore DB01 chain and FS01 sample to an isolated network; run database consistency checks, row counts, latest-transaction time and owner sample checks	Monthly (DB01); quarterly (others)	Restore time, achieved RPO, owner sign-off
Malware screening	Scan restored images before connection; compare admin accounts with approved list	Every restore	Scan report

Task 3 Recovery strategy and budget

- Compare the Cold, Warm, and Hot packages against your RTO and RPO targets, staffing needs, geography, supplier dependencies, and the risk of propagating corruption or malicious changes. Treat the quoted recovery windows as planning estimates.

ANSWER

Table 10. Option matrix against objectives and constraints

Criterion	Cold	Warm	Hot
First critical service (estimate)	24 to 48 h	3 to 5 h	About 1 h
F1a examination (MTD 3 h)	Not met	Not met without continuity workaround	Met if estimate holds
F1b, F2, F3 RTOs (4 h, 5 h, 6 h)	Not met	Met at 3 to 4 h; at risk at 5 h	Met
F4, F5 RTOs (24 h)	At risk (24 to 48 h)	Met	Met
Data-loss window vs 1 h RPO	Up to 24 h: not met	1 to 6 h: met only at best case; protected backups (Table 7) needed	Up to 15 min: met
Staffing	Full rebuild; two technicians insufficient for 72 h	Restoration by two technicians (Task 5 shows 4 h 30 for all)	Replication monitoring skills; less restore effort
Geography	Must be outside the flood area; location unknown	Same requirement; confirm flood exposure	Often near the primary for low latency, which may share flood and grid risk
Supplier dependency	Hardware delivery under pressure	Hosting supplier and backup provider	Highest; continuous replication link and supplier

Corruption or malicious change	Restores from backups; can choose a clean point	Restores from versioned backups; can choose a clean point	Replication can copy corruption or malicious changes within minutes

- Calculate the complete annual cost of each package, including both required additions. Identify which options fit the NGN 3,000,000 limit. Recommend one option, justify the tradeoffs, and identify any objective it cannot meet. Document residual risks and who must accept them.

ANSWER

Table 11. Complete annual cost of each package

Cost item (NGN per year)	Cold	Warm	Hot
Annual base package	600,000	1,500,000	2,400,000
Independent protected backups	400,000	400,000	400,000
Training and tests	200,000	200,000	200,000
Complete annual cost	1,200,000	2,100,000	3 ,000,000
Planning limit	3,000,000	3,000,000	3,000,000
Headroom (limit − cost)	1,800,000	900,000	0
Within limit?	Yes	Yes	Yes, exactly at limit; no contingency

Table 12. Residual risks of the Warm recommendation and risk acceptance

Residual risk	Treatment	Accepted by
Examination at 11:00 cannot rely on technical recovery	Pre-approved paper or M365 alternative; reschedule as last resort	Head of Academic Services, on behalf of the Executive Steering Committee
Warm timing may be at 5 h, not 3 h	Validate in annual parallel test before contract renewal	Director of Operations
Data-loss window up to 6 h if package performs at worst case	Protected 30-minute DB01 backups and reconciliation procedures	Finance Manager (F2) and Director of Operations
Same-region flood or grid failure affecting warm site	Require flood-exposure statement for the warm site	Principal (Executive Sponsor)
Supplier recovery claims unproven	Evidence list below before signing	Executive Steering Committee

- Compare your recommendation with the estimated 8-hour and 24-hour disruption costs. Explain why a single-outage estimate cannot by itself establish annual return on investment. Identify the evidence needed before approving expenditure or accepting supplier recovery claims.

ANSWER

Table 13. Illustrative break-even comparison (Modelled baseline result, not a live test)

Item	Working	Result (NGN)
Impact per event, Cold (no service within 24 h)	Full 24-hour impact	3,360,000
Impact per event, Warm (Task 5 baseline)	F1 3.75 × 60,000 = 225,000; F2 3.5 × 40,000 = 140,000; F3 4 × 20,000 = 80,000; F4 4.5 × 10,000 = 45,000; F5 2.5 × 5,000 = 12,500; + 120,000 rescheduling	622,500
Impact per event, Hot (about 1 h, no rescheduling)	135,000 × 1	135,000
Warm over Cold: break-even event frequency	900,000 extra cost ÷ (3,360,000 − 622,500 = 2,737,500) avoided	0.33 events per year
Hot over Warm: break-even event frequency	900,000 extra cost ÷ (622,500 − 135,000 = 487,500) avoided	1.85 events per year

A single-outage estimate cannot by itself establish annual return on investment. ROI depends on how often site-loss events occur and how severe they are, and neither frequency nor probability is known. The package also protects against other disruptions such as hardware failure and ransomware. The comparison shows only that Warm pays back if a comparable event happens about once every three years, while Hot needs nearly two events a year to justify its extra cost. This is consistent with the deck's financial metrics, which treat cost of downtime avoided and ROI as measures to track over time.

Before approving expenditure or accepting supplier claims, the Executive Steering Committee should require: incident and flood history for the main and warm sites; independent test reports showing achieved recovery times and data points; contract terms stating RTO, RPO, capacity for all five functions and service credits; a documented description of the backup provider's access independence; and a Northbridge-run parallel test of the warm site before renewal (Swanson et al., 2010).

Coordinated continuity and recovery plan

Task 4 Business continuity and disaster recovery plan

- Prepare a coordinated plan with scope, activation thresholds, an authorised decision maker and deputy, escalation, contact roles, alternative communication, and situation reporting. Define how the continuity team, technical recovery team, facilities staff, and incident response team work together.

ANSWER

Scope, activation and decision authority

Table 14. Plan scope, activation thresholds and decision authority

Element	Northbridge plan content
Scope	F1 to F5; systems LMS01, FIN01, REG01, HR01, DB01, AUTH01, FS01 and the recovery network; main site, alternate facility and warm recovery site; all 90 staff and 1,800 learners. Microsoft 365 is out of DR scope but used for continuity.
Activation thresholds	**Level 1 Alert:** disruption to the site or a shared system expected to last under 2 hours; BC/DR Coordinator monitors. **Level 2 Partial:** one function at risk of missing its RTO, or loss of DB01 or AUTH01 for over 1 hour in working hours. **Level 3 Full:** main site unavailable for over 4 hours, or any threat to an examination within 4 hours. Day 0 meets Level 3 at 08:00.
Decision maker	**Incident Commander: Director of Operations.** Deputy: Deputy Principal (Academic). If neither answers within 15 minutes, the BC/DR Coordinator may activate Level 2 and must escalate.
Escalation	To the Principal (Executive Sponsor) for Level 3, outage over 24 hours, unbudgeted spend over NGN 500,000, or suspected compromise. To the Executive Steering Committee for risk acceptance.
Contact roles	Communications Lead, Facilities Manager, IT Manager (Technical Recovery Lead), Information Security Officer (IR Lead), Data Protection Officer, function owners. Named contacts and alternates sit in an offline annex (fictional).
Alternative communication	Primary: M365 Teams and email over mobile internet. Backup: SMS broadcast and phone tree. Last resort: printed contact cards held by each role holder.
Situation reporting	SITREP template (status per function, decisions, risks, next actions) at 08:45, 10:00, 11:30, 13:00, 16:30 on Day 0, then 09:00 and 16:00 daily, published in the M365 BC channel.

Table 15. Team structure and coordination

Team (slide 12)	Northbridge members	Role during the outage	Works with
Crisis Management Team	Incident Commander (Director of Operations), Communications Lead, HR Manager, Facilities Manager, external Legal Counsel	Strategic decisions, approvals, spending and communications	All teams through SITREPs every 90 to 210 minutes
Continuity team (BC coordinators)	BC/DR Coordinator (programme manager), function owners for F1 to F5, Examinations Officer, Learner Support Coordinator	Runs temporary procedures and keeps manual records	DR team for acceptance tests; Facilities for alternate site
Technical recovery (DR) team	IT Manager, technicians T1 and T2	Runbook execution, validation, rollback	IR team before any data is trusted

Facilities	Facilities Manager, alternate facility provider	Site safety, alternate facility, return clearance	Crisis team; DR team for power and network
Incident response	Information Security Officer, Data Protection Officer	Joins if compromise is suspected (Section 4.6)	DR team; Crisis team on notification

- Create a RACI matrix for at least six activities, including activation, temporary assessment delivery, backup validation, restoration, stakeholder communication, and return to the primary site. Give each activity exactly one accountable role and at least one responsible role. A combined A/R role is acceptable.

ANSWER

RACI matrix

Table 16. RACI matrix (exactly one A per activity)

Activity	Inc. Cmdr	BC/DR Coord.	IT Mgr	T1 / T2	Function owners	Comms Lead	Facilities Mgr	IR Lead
Plan activation	A	R	C	I	C	I	C	I
Temporary assessment delivery	C	C	C	I	A/R (Head of Academic Services)	C	R	I
Backup selection and validation	I	I	A	R	I			C
System restoration	I	C	A	R	C			C
Business acceptance of each function	I	R	C	C	A/R			
Stakeholder communication	A	C	C		C	R	C	C
Incident response engagement	I	C	R	C		I		A/R
Return to the primary site	A	R	R	R	C	I	R	C

Inc. Cmdr: Incident Commander (Director of Operations); BC/DR Coord.: BC/DR Coordinator; IT Mgr: IT Manager; Facilities Mgr: Facilities Manager. R responsible, A accountable, C consulted, I informed. A/R means the accountable role also performs the work. For acceptance, each function owner is accountable for their own function only.

- Design temporary procedures for F1 to F5 during the 72-hour site outage. Address the 25-workstation limit and 40 learners needing a centre workstation. Explain approvals, accessibility, assessment

integrity, privacy, record keeping, and later reconciliation. Identify the minimum staff, equipment, and supplier support required.

ANSWER

Table 17. Examination continuity allocation for the 11:00 sitting

Group	Candidates	Delivery	Integrity and accessibility controls
Own devices	80	M365 Forms examination, released at 11:00 with Teams camera invigilation	Single-use link; time window; identity check against M365 account; invigilator log
Workstations	25	Same Forms examination at the alternate facility	Seat plan; locked-down browser profile prepared by provider (A11); invigilators
Paper	15	Printed equivalent paper in a second invigilated room	Papers printed at 10:30 under Examinations Officer control; numbered scripts; signed register
Total	120	All start at 11:00	Candidates unable to travel safely are rescheduled without penalty

Table 18. Temporary procedures, approvals and reconciliation

Function	Temporary procedure (Day 0 to Day 3)	Approval and record controls	Privacy, accessibility and reconciliation
F1	Examination as in Table 17. Teaching moves to Teams with recorded sessions. Coursework deadlines extended by 72 hours.	Head of Academic Services approves the alternative by 09:00; Examinations Officer keeps registers and script log.	Recordings in restricted Teams channel. Results entered into LMS01 after restoration and checked against scripts by two staff.
F2	Payments continue through the provider. Provisional receipts issued from a restricted M365 workbook with sequential numbers. No manual wallet changes.	Finance officer prepares; Finance Manager approves each batch. Workbook version history kept.	Workbook limited to finance staff. Every entry matched to provider transaction ID after FIN01 is restored.
F3	Applications and changes captured on an M365 form until 18:00, with automatic reference and time stamp. No offers issued until entered in REG01.	Admissions Manager reviews daily; any deadline extension needs Registrar approval.	Form accessible on mobile; data collected is the minimum for application. Entries keyed into REG01 with second-person check.
F4	If HR01 is not available by 08:00 Day 1, submit the last approved payroll with approved variances only.	HR Manager and Finance Manager dual authorisation; unique batch reference; bank confirmation checked before any resubmission.	Payroll file kept encrypted. Variances corrected in the next run and recorded.
F5	Requests logged in a shared mailbox with reference and 48-hour due time.	Registry Manager approves each issue; issue log kept.	No certificates generated from unverified copies. Log

	Certificates issued only after FS01 is verified.		reconciled with FS01 after return.

Table 19. Minimum staff, equipment and supplier support

Resource	Minimum requirement
Staff	Incident Commander or deputy; BC/DR Coordinator; Examinations Officer plus six invigilators; one finance preparer and one approver; one admissions officer; one HR officer; one registry officer; Communications Lead; two learner support staff; T1 and T2 with the IT Manager.
Equipment	25 alternate workstations; printer and secure paper store; staff laptops with mobile internet (A15); power banks.
Supplier support	Alternate facility provider (workstations, rooms, power); warm-site host; offsite backup provider for retrieval; payment provider reports (A14); bank payroll helpdesk.

- Write a DR runbook with ordered actions, prerequisites, owners, completion checks, and contingency actions if a step fails. Include backup selection, validation, application and data checks, business acceptance, and rollback. Distinguish restoring technology from resuming an acceptable business service.

ANSWER

Table 20. Disaster recovery runbook

Step	Action	Prereq.	Owner	Completion check	Contingency if the step fails
R1	Declare DR; freeze changes; confirm no compromise indicators	Level 3 activation	IT Manager	Logged decision	Engage IR Lead (4.6)
R2	Recovery network	R1	T1	Isolated segment up; firewall rules applied	Use provider standby router; escalate to host
R3	Backup selection: B03, B04, B01 + B02 chain; record catalogue IDs	R1	IT Manager	Selection signed	Disconnected full (Day −3)
R4	AUTH01 restoration from B03	R2, R3	T1	Test logons; admin accounts match approved list	Rebuild from Day −3 copy; reset privileged passwords
R5	Security and backup checks	R4	T1	Hashes match; chain complete; malware scan clean; backup logon audit reviewed	Stop and engage IR; restore to last good link

R6	DB01 restoration (B01, then each B02 file to 07:00)	R5	T1	Consistency check passes; latest transaction 07:00	Restore to last valid incremental; record new RPO
R7	FS01 restoration from B04	R4, R5	T2	Checksums; access restrictions; sample files open	Day −3 copy; re-create files from email
R8	LMS01, FIN01, REG01, HR01	R6	T1 / T2	Application test transaction succeeds	Reinstall from image; keep manual procedure
R9	Pre-acceptance snapshot of each restored system	R6 to R8	T2	Snapshot ID logged	Delay acceptance until snapshot exists
R10	Business acceptance against Table 5 declaration level	R9	Function owner	Signed acceptance	Rollback (R11)
R11	Rollback: revert to snapshot or earlier checkpoint; users stay on temporary procedure	Failed R10	IT Manager	Service on prior state; decision logged	Escalate to Incident Commander
R12	Switch users; reconcile temporary records; monitor	R10	BC/DR Coordinator	Reconciliation report per function	Extend temporary procedure

- Provide a stakeholder communication matrix and a 120 to 180 word initial learner notice. Explain when possible compromise would trigger incident response involvement. Define safe return to the primary site, controlled transfer of new records, and final sign-off after facilities clearance and business validation.

ANSWER

Subject: Main Site Closed Today – Please Read Before Travelling

Dear Learner,

Please be informed that our main site is closed from this morning due to flooding and a power failure. The closure is expected to last approximately three days. **Please do not travel to the main site.** No cyberattack has been identified.

Today's 11:00 examination will proceed as scheduled. If you have been instructed to use your own device, please sign in to your Northbridge Microsoft 365 account by 10:30 and follow the instructions sent by email.

If you require a centre workstation, we will text you the room and seat details at the alternate facility by 10:00. If you cannot travel safely or require an adjustment, please reply to this message. You will not be penalised.

Classes will move to Microsoft Teams today. Fee payments through our payment provider remain available, although receipts may be delayed. Admissions are still open, and applications can be submitted through the online form until 18:00.

Our next update will be provided at 11:30 today.

Thank you for your understanding and cooperation.

Kind regards,

Northbridge Continuity Team.

Table 21. Stakeholder communication matrix

Stakeholder	Message	Channel	Sender	Timing	Approval
Learners	Initial notice (below); updates	SMS, email, LMS banner when restored	Communications Lead	08:45, then each SITREP	Incident Commander
Examination candidates	Seat, room or link and time	SMS and email	Examinations Officer	By 10:00	Head of Academic Services
Staff	Work location, duties, safety	Teams, SMS phone tree	HR Manager	08:30, then each SITREP	Incident Commander
Executive Steering Committee	SITREP, decisions, risk acceptance	Teams call, email	BC/DR Coordinator	Each SITREP	Incident Commander
Suppliers	Activation and requirements	Phone, then email	IT Manager; Facilities Manager	08:15	Incident Commander
Payment provider and bank	Report requests; payroll timing	Phone, email	Finance Manager	09:00	Finance Manager
Awarding partners and sponsors	Examination arrangements	Email	Head of Academic Services	By 12:00	Incident Commander
Media and public	Holding statement if asked	Website, social media	Communications Lead	On request	Principal

No cyberattack has been confirmed, so this is a BC/DR event. The deck notes that major cyber incidents activate both disciplines and need unified command (ICDFA, n.d., slide 14). The IR Lead is engaged immediately, and restoration of the affected component pauses, if any of the following occurs: a backup

fails its integrity check without a clear technical cause; an unexpected privileged or backup administrator logon; unknown accounts or changes in restored AUTH01; malware found in a restored image; or business data that differs from external records in a way the data-loss window does not explain. The IR Lead then decides whether to preserve evidence, isolate the affected copy and change the restore point. The Incident Commander keeps overall authority.

Table 22. Return to the primary site and final sign-off

Stage	Criteria	Sign-off
1 Facilities clearance	Water removed; electrical safety inspection passed; power stable for 24 hours; environmental and structural checks; flood-damaged equipment quarantined	Facilities Manager
2 Technical readiness	Primary hardware tested; network and AUTH01 rebuilt or verified; backups running to the new schedule	IT Manager
3 Controlled transfer of new records	Change freeze at the recovery site; final backup and hash; transfer to primary; record counts and hashes compared; temporary records reconciled	IT Manager with function owners
4 Business validation	Each owner repeats the Table 5 declaration test on the primary site; parallel running for one business day	Function owners
5 Final sign-off	Recovery site kept on standby for 7 days; plan stood down; review scheduled within 10 working days	Incident Commander

Recovery validation and program governance

Task 5 Recovery schedule and exercise

- Using the current warm site timings, produce a schedule with action, technician, start time, finish time, and prerequisite. Respect the two-technician limit. Determine the earliest possible restoration of assessment service and a feasible completion time for all five functions. Show the dependency path that controls the assessment finish time.

ANSWER

Baseline schedule

The schedule in Table 23 uses the current warm-site drill timings, starts at 08:00, gives each action to one technician for its full duration, never splits an action and starts an action only when all its prerequisites are complete. It is a **Modelled baseline result, not a live test**. It assumes B01 to B04 pass their integrity checks, as the brief specifies for the baseline drill.

Table 23. Baseline recovery schedule, two technicians (Modelled baseline result, not a live test)

Action	Technician	Start	Finish	Minutes	Prerequisite
Recovery network	T1	08:00	08:30	30	None

AUTH01 restoration	T1	08:30	09:00	30	Recovery network
Security and backup checks	T1	09:00	09:30	30	AUTH01
DB01 restoration	T1	09:30	11:00	90	Security and backup checks
FS01 restoration	T2	09:30	10:30	60	AUTH01 and checks
FIN01 restoration	T2	11:00	11:30	30	DB01
LMS01 restoration	T1	11:00	11:45	45	DB01
REG01 restoration	T2	11:30	12:00	30	DB01
HR01 restoration	T1	11:45	12:30	45	DB01

Table 24. Schedule chart in 15-minute slots, hour marks 08 to 12 (dark T1, light T2)

Action	08				09				10				11				12	
T1 Network																		
T1 AUTH01																		
T1 Checks																		
T1 DB01																		
T2 FS01																		
T2 FIN01																		
T1 LMS01																		
T2 REG01																		
T1 HR01																		

The path that controls the assessment finish is recovery network (30) → AUTH01 (30) → security and backup checks (30) → DB01 (90) → LMS01 (45), a total of 225 minutes. Each step needs the one before, so no allocation of technicians can shorten it. Assessment service is therefore available at 11:45 at the earliest, and the schedule achieves that bound. All five functions are complete by 12:30. I checked whether this is optimal for total completion. The four application actions after DB01 total 150 minutes, so two technicians need at least 75 minutes after 11:00, and 12:15 is the earliest possible finish (T1: LMS01 then REG01; T2: FIN01 then HR01). I keep the 12:30 schedule because it restores F3 before F4 in business-priority order and frees T2 at 12:00 to support finance reconciliation. Both are feasible.

Table 25. Elapsed recovery time and checkpoints against targets (Modelled baseline result, not a live test)

Function	Ready	Elapsed	RTO	MTD	RTO result	Restored checkpoint (age)	RPO	RPO result
F1 (LMS01)	11:45	3 h 45	4 h (F1b); 3 h (F1a)	8 h; 3 h	F1b met (15 min spare); F1a missed	DB01 07:00 (1 h)	1 h	Met, no margin
F2 (FIN01)	11:30	3 h 30	5 h	8 h	Met	DB01 07:00 (1 h)	1 h	Met, no margin
F3 (REG01)	12:00	4 h 00	6 h	10 h	Met	DB01 07:00 (1 h)	2 h	Met
F4 (HR01)	12:30	4 h 30	24 h	28 h	Met	DB01 07:00 (1 h)	4 h	Met
F5 (FS01)	10:30	2 h 30	24 h	48 h	Met	FS01 23:00 Day −1 (9 h)	4 h	Not met (5 h gap)
All (AUTH01)	09:00	1 h 00	n/a	n/a	n/a	AUTH01 00:00 (8 h)	4 h	Not met (4 h gap)

Even with successful restoration, several deadlines still need a continuity workaround. The 11:00 examination falls 45 minutes before LMS01 is available, so the alternative in Table 17 is essential. Fee confirmation at 16:00 is technically possible from 11:30, but payments made after 07:00 are missing from FIN01 and must be reconciled from provider reports. Applications received between 08:00 and 12:00 must be keyed from the form. Payroll is restored well before 12:00 on Day 1, but HR changes made after the checkpoint need checking. FS01 files from the 9-hour gap must be re-created from request emails.

- Calculate each function's elapsed recovery time from 08:00 and compare it with your RTO and MTD targets. Compare restored checkpoints with your RPO targets. Label these as modelled baseline results, not a live test. Explain which deadlines need a continuity workaround even when restoration is successful.

Design and walk through a 45 minute tabletop exercise using these injects. You may simulate all participant roles individually.

Exercise time	Required inject
Minute 10	The site closure extends from 72 hours to five days.
Minute 20	The alternate facility can provide only 15 working workstations.
Minute 30	A DB01 incremental backup fails its integrity check. An unexpected backup administrator logon also requires investigation.
Minute 40	The payment provider reports successful payments absent from the restored finance records.

ANSWER

Tabletop exercise design

Table 26. Tabletop exercise design

Element	Design
Type and length	Discussion-based tabletop, 45 minutes, no system changes (ICDFA, n.d., slide 11). I simulated every role individually.
Objectives	Test decision authority, examination continuity, backup integrity response and finance reconciliation under stress.
Roles simulated	Incident Commander, BC/DR Coordinator (facilitator), IT Manager, T1, T2, Head of Academic Services, Finance Manager, IR Lead, Communications Lead, Facilities Manager.
Timeline	0 to 10 min: baseline brief at 08:00 Day 0; injects at minutes 10, 20, 30 and 40; 40 to 45 min: inject 4 and hot debrief.
Success criteria	A decision with an owner within 5 minutes of each inject; no data trusted before validation; IR engaged when triggers in 4.6 occur; learners informed at the next SITREP.

Table 27. Inject log with decisions (Simulated tabletop result)

Minute	Inject	Decision	Revised action	Owner	Rationale	Effect on service or recovery
10	Site closure extends from 72 hours to five days	Extend Level 3 activation to Day 5; plan for a full working week away from site	Extend alternate facility and warm-site booking; two-shift rota for T1, T2 and coordinators; reissue learner notice; schedule week's examinations at the alternate facility	Incident Commander	Temporary procedures were sized for 72 hours; staff fatigue and supplier capacity become risks	F4 payroll and F5 48-hour commitment now fall wholly inside the outage; extra hire cost escalated to Principal
20	Alternate facility has only 15 working workstations	Keep one 11:00 sitting: 15 on workstations, 25 on invigilated paper	Allocate workstations by access need first; print 25 papers; open a third room; reschedule only those unable to attend	Head of Academic Services	Single sitting protects paper security; paper is an accepted equivalent (A23)	Rescheduling cost avoided if accepted; more invigilators needed (from 6 to 8)
30	A DB01 incremental fails its integrity check, and an unexpected backup	Treat as possible compromise: engage IR; suspend the shared admin identity;	IR Lead reviews logon and job logs (assumed 60 minutes, A18); T1 restores B01 plus incrementals to 03:00, the last link before the failed	IR Lead (investigation); IT Manager (restore)	Later incrementals depend on the failed link; restoring possibly altered data	DB01 RPO rises from 1 h to 5 h, breaching the 1 h target; DB01 restore runs 10:30 to 12:00, LMS01 by 12:45,

	administrator logon needs investigation	restore DB01 only from validated links into an isolated segment	04:00 file; Day −3 copy prepared in case B01 is suspect		could spread harm	FIN01 by 12:30, REG01 by 13:00, HR01 by 13:30
40	Payment provider reports successful payments absent from restored finance records	Provider records are authoritative for payment status; no learner marked unpaid	Match every provider transaction ID to FIN01; post missing items with dual approval; flag any absent payment older than the restore point to IR; issue 16:00 confirmation from reconciled data	Finance Manager	Gap is expected for payments after 03:00; any older gap may indicate tampering	F2 declared recovered only after reconciliation, about 15:00; 16:00 deadline still met

These outcomes are a **Simulated tabletop result** and are kept separate from the baseline in Table 23. The main lesson is that the baseline depended on one assumption, that the backup chain was sound, which the shared administrator identity made hard to trust. The exercise also showed that the examination and fee deadlines can be protected by continuity procedures even when technical recovery slips by an hour.

- Record a decision and revised action for every inject, including its owner, rationale, and effect on service or recovery. Keep this stress scenario separate from the baseline schedule. Produce at least five corrective actions with priority, owner, target date, and measurable completion evidence.

ANSWER

Table 28. Corrective action plan (target dates assume approval in November 2026, A21)

ID	Corrective action	Priority	Owner	Target date	Measurable completion evidence
CA1	Separate production and backup administrator identities with MFA and a sealed break-glass account	High	IT Manager	30 Nov 2026	Access review shows zero shared admin accounts; logon alert test recorded
CA2	Obtain documented access-independence controls from the offsite provider and add them to the contract	High	Director of Operations	15 Dec 2026	Signed contract annex; failed deletion test with production credentials
CA3	Automated chain integrity checks with alerting for every DB01 backup job	High	IT Manager	30 Nov 2026	30 consecutive daily reports; test alert received within 15 minutes
CA4	Implement Table 7 schedule for DB01, AUTH01 and FS01	High	IT Manager	31 Jan 2027	Restore test achieving 30 min, 4 h and 2 h checkpoints
CA5	Examination continuity pack: pre-approved paper and Forms versions, workstation allocation rules	High	Head of Academic Services	15 Jan 2027	Pack approved by Academic Board; used in one walkthrough

CA6	Finance reconciliation procedure using provider transaction IDs	Medium	Finance Manager	31 Jan 2027	Procedure approved; reconciliation of a test day with 100% match
CA7	Alternate facility agreement for 40 workstations and up to 5 days	Medium	Facilities Manager	28 Feb 2027	Signed agreement stating capacity and duration

Task 6 Governance and performance measures

- Assign plan ownership and approval. Propose a risk-based testing calendar with review triggers, training, version control, and distribution. Map at least four of your outputs to NIST SP 800-34 planning steps and four to the ISO 22301 themes covered in the slides. Explain one relevant flood or climate consideration.

ANSWER

Plan Ownership, Testing Calendar, Mapping, and Climate Consideration

Ownership and Approval

The Business Continuity Plan (BCP) is owned by the Business Continuity Manager (or equivalent role in the risk/compliance function), who is responsible for content accuracy, updates, and day-to-day maintenance. Approval authority rests with the Executive Sponsor (e.g., Chief Risk Officer or Head of Operations) after review by the Business Continuity Steering Committee. Final sign-off is required from senior leadership before any major version is released.

Risk-Based Testing Calendar

Testing frequency is driven by residual risk level and criticality of the process/function:

- High-risk / critical processes (e.g., those with RTO $\leq$ 4 hours): tabletop exercises quarterly and full restoration or failover tests at least twice per year.
- Medium-risk processes: tabletop exercises twice yearly and one technical recovery test annually.
- Low-risk processes: annual tabletop exercise.

Review Triggers

Mandatory reviews occur after any of the following: actual disruption or near-miss, significant organizational or technology change, failed test or audit finding, regulatory update, or at least every 12 months (whichever comes first).

Training

All process owners and recovery team members complete role-based training at least annually, with refresher sessions after major plan updates. New staff receive induction training within 30 days of joining a recovery role.

Version Control and Distribution

All plan versions are controlled through a document management system with unique version numbers,

change logs, and approval records. The current approved version is distributed only to authorized recovery team members via secure channels. Superseded versions are archived and marked obsolete.

Mapping to NIST SP 800-34 Planning Steps (at least four)

1. Conduct Business Impact Analysis → Identification of critical functions, RTO, RPO, and impact ratings.
2. Identify Preventive Controls → Risk treatments and controls linked to the BIA findings.
3. Develop Contingency Strategies → Recovery strategies and resource requirements.
4. Ensure Plan Testing, Training, and Exercises → The risk-based testing calendar and training programme above.

Mapping to ISO 22301 Themes (at least four)

1. Leadership → Plan ownership, executive approval, and steering committee oversight.
2. Planning → Risk-based testing calendar and review triggers.
3. Support → Training and competence requirements.
4. Performance Evaluation → KPIs (covered in the next section) and post-test reviews.

Flood / Climate Consideration
Given the location's exposure to seasonal flooding (common in the Port Harcourt / Rivers State area), the BIA and recovery strategies explicitly address flood risk. Critical IT and operational sites must have elevated equipment or secondary locations outside known flood zones, with recovery strategies that include alternative power, data centre failover, and physical access contingencies for periods when roads or facilities are inundated. This is reviewed annually against updated climate or flood-risk data.

- Define at least four KPIs with formula or measurement method, data source, target, reporting frequency, and accountable owner. Include recovery performance, restoration-test coverage, corrective-action closure, and training. Assess current maturity using the slide 13 model and propose a realistic 12-month improvement target.

ANSWER

KPIs, Maturity Assessment, and 12-Month Improvement Target

Four Key Performance Indicators

The KPIs follow the five categories on slide 13 and include the four that the brief requires: recovery performance, restoration-test coverage, corrective-action closure and training (ICDFA, n.d., slide 13). The current values come only from the modelled baseline and the tabletop.

Table 29. KPI dashboard by slide 13 category

Category (slide 13)	KPI	Formula or method	Data source	Target	Freq.	Owner	Current value

Recovery Performance	K1 Functions recovered within RTO	Functions meeting RTO in test ÷ functions tested × 100	Restore and parallel test records	100% for F1b to F3; 100% overall	Each test; quarterly report	IT Manager	5 of 5 modelled (100%); F1a needs workaround
Recovery Performance	K2 Achieved RPO	Checkpoint age at test start vs RPO per system	Backup catalogue, integrity report	All systems within RPO	Monthly	IT Manager	1 of 3 systems (DB01 only)
Testing & Exercises	K3 Restoration-test coverage	In-scope systems with a successful restore test in the quarter ÷ 7 × 100	Test log	100% per quarter	Quarterly	BC/DR Coordinator	Not measured; baseline drill only
Testing & Exercises	K4 Corrective-action closure	Actions closed with evidence by target date ÷ actions due × 100	Corrective action register	≥ 90%	Monthly	BC/DR Coordinator	0 of 7 (none yet due)
Compliance & Governance	K5 Training completion	Role holders and deputies trained in last 12 months ÷ total × 100	HR training records	100% primary; ≥ 90% deputies	Quarterly	HR Manager	Not measured
Availability & Reliability	K6 Mean time to restore critical service	Average minutes from disruption to F1b to F3 declaration	Incident and test logs	≤ 240 minutes	Per event; annual	IT Manager	225 minutes for F1b (modelled)
Financial Metrics	K7 Cost of downtime avoided	Modelled no-recovery impact minus actual impact, per event	BIA rates, event log	Report trend	Annual	Finance Manager	NGN 2,737,500 (modelled)

Using the slide 13 maturity model, I assess Northbridge at **Level 2: Developing (Repeatable)**. Basic recovery capability exists for critical systems: offsite backups, a drill with measured timings and an alternate facility. However, testing has not been regular, backup administration is not independent, provider controls are undocumented and there is no approved plan or KPI reporting (ICDFA, n.d., slide 13; A22). Level 1 would understate the evidence, and Level 3 requires documented processes, regular testing and clear governance, which are not yet in place.

Table 30. Maturity assessment using the slide 13 BC/DR Maturity Assessment Model

Level (slide 13)	Description from the deck	Northbridge evidence
1 Initial (Ad Hoc)	Reactive, informal, no formal plans	Exceeded: backups and drill timings exist

2 Developing (Repeatable)	Basic plans for critical systems; some testing, not regular	**Current level.** Drill and backups exist; gaps in independence and testing
3 Defined (Standardized)	Documented processes, regular testing, clear governance	**12-month target (by Oct 2027).** Approved plan; calendar delivered; KPIs K1 to K5 reported quarterly
4 Managed (Measured)	Data-driven, proactive gap identification, linked to enterprise risk	Longer-term aim once K1 to K7 have a year of data
5 Optimizing	BC/DR a core competency, embedded in culture	Not realistic within 12 months

Current Maturity Assessment

Using the maturity model presented on slide 13 (typically a five-level scale from Initial/Ad-hoc to Optimized), the organisation is currently assessed at **Level 2 – Repeatable**. Core processes exist and some testing and documentation occur, but they are not yet fully standardised, consistently measured, or embedded across all critical functions. Governance and KPIs are emerging but not systematically applied.

Realistic 12-Month Improvement Target

Move from Level 2 (Repeatable) to **Level 3 – Defined** within 12 months. This will be achieved by fully implementing the ownership model, testing calendar, version control, and the four KPIs above, with at least one full cycle of risk-based testing completed, all critical process owners trained, and regular reporting of the KPIs to the steering committee.

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